

Federico Bertoia

Ph.D. Candidate in Statistics, Ghent University

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RESEARCH INTERESTS

Causal Inference, Debiased Machine Learning

EDUCATION

Ghent University, Ghent, Belgium 2024 – Present
Ph.D. in Statistics
Supervisor: Prof. Dr. Stijn Vansteelandt

University of Bologna, Bologna, Italy 2021 – 2023
M.Sc. in Economics and Econometrics, with honours

University of Bologna, Bologna, Italy 2018 – 2021
B.Sc. in Economia Aziendale, with honours

EXPERIENCE

Junior Data Scientist, CRIF 2024

- Contributed to the development of a Generative AI framework for analyzing PDF documents, including a personalized chatbot, KPI extraction, and document summarization.

Research Assistant – MinUTo Project, University of Bologna 2024

- The project studies how parenting practices influence children's non-cognitive skills and social preferences using an RCT. Worked on the empirical analysis, focusing on instrumental-variables estimation and heterogeneous treatment effects.

Research Intern, ISTAT - Italian National Institute of Statistics 2023

- Creation of supervised machine learning models for Natural Language Processing. The task involved using state-of-the-art Transformer models (BERT) for word embedding and a neural network for the classification of textual documents.

TEACHING EXPERIENCE

Teaching Assistant – Causal Machine Learning, Ghent University 2025 - Present

WORK IN PROGRESS

- *Debiased Risk Minimization with Application to Conditional Average Shift Intervention Effects* with Achille Demares, Georgi Baklcharov and Stijn Vansteelandt.
- *The toll of an early bell? School start time and adolescent sleep quality in Flanders, Belgium* with Ann Vandendriessche, Louise Poppe, Thijs Galmart, Margaux Rigole, Stijn Vansteelandt and Maxim Dierckens.

CONFERENCE PRESENTATIONS

- Isaac Newton Institute Workshop on Causality and Machine Learning (poster) - Cambridge, United Kingdom (2026)
- European Causal Inference Meeting - Oxford, United Kingdom (2026)
- International Conference on Statistics and Data Science - Seville, Spain (2025)

TECHNICAL SKILLS

R, Python, $\text{\LaTeX}$

LANGUAGES

Italian (native), English (advanced)